KENTUCKY PORTABLE CLUB • FIELD RESOURCE

ANTENNA EFFICIENCY AND OPERATING DURING THE SOLAR MINIMUM

Practical guidance for portable operators who want to get the most from every watt, every band, and every activation.

GET OUTSIDE. GET ON THE AIR.

Antenna Efficiency and the Solar Minimum

When solar activity drops toward a **solar minimum**, many amateur radio operators notice a change in the way the bands behave. But lower solar activity does not mean your antenna becomes less efficient. Instead, it makes **antenna efficiency and good station setup even more important.**

Solar minimum conditions generally mean weaker ionospheric support on the higher HF bands, particularly 10 and 12 meters. Lower-frequency bands such as 20, 30, 40, and 80 meters often become more valuable for reliable long-distance communication.

Why Antenna Efficiency Matters

When propagation is weak, every decibel counts. A well-built, properly installed antenna can put more of your transmitter's power into the radiating system and give your receiver a better chance of hearing weak signals.

Efficiency is affected by more than just the antenna itself. **Feed-line loss, poor connections, inadequate counterpoises, ground losses, antenna height, nearby objects, and poor tuning** can all reduce performance.

A useful way to understand efficiency is to compare transmitter power with the power that is actually radiated. For example, if a station delivers 100 W to an antenna system that is 70% efficient, only about 70 W is radiated and approximately 30 W is lost as heat or other system losses.

SYSTEM EFFICIENCY	RADIATED FROM 100 W	POWER LOST
95%	95 W	5 W
90%	90 W	10 W
80%	80 W	20 W
70%	**70 W**	**30 W**
60%	60 W	40 W
50%	50 W	50 W
25%	25 W	75 W

A low SWR reading alone does not guarantee high efficiency. A tuner can make a lossy antenna look electrically matched while significant power is still being dissipated in ground losses, loading coils, resistive connections, or feed-line loss.

Operating Modes and Antenna Efficiency

Antenna efficiency matters regardless of the operating mode, but different modes place different demands on the station.

SSB

Useful for conversational contacts and nets on 20, 40, and 80 meters. Good microphone technique, proper audio levels, and avoiding excessive compression are important when signals are weak.

CW

CW can provide reliable communication with relatively low power because a continuous carrier and narrow bandwidth let the receiver concentrate on a small portion of the spectrum.

FT8 and Other Digital Modes

Digital modes such as FT8 can decode signals that may be too weak for comfortable voice communication. They can help operators work weak signals, but they are not a substitute for an efficient antenna. Improving antenna height, feed-line loss, grounding, and counterpoise performance benefits digital modes just as it benefits SSB and CW.

Choosing the Mode

- **SSB** is useful for conversational contacts and nets.
- **CW** is highly effective when signals are weak and bandwidth needs to be minimized.
- **FT8** and similar digital modes can reveal openings and complete contacts at very low signal levels.
- Other digital modes may offer better performance for messaging, weak-signal work, or longer exchanges.

> **The important point is that an efficient antenna improves every mode. A strong, clean signal and a sensitive receiving system give the operator more options when conditions change.**

Some Highly Efficient Antenna Choices

FULL-SIZE DIPOLE

A properly installed full-size half-wave dipole remains one of the simplest and most efficient HF antennas available. For portable operations, an inverted-V can provide an excellent combination of efficiency, simplicity, and portability.

END-FED HALF-WAVE (EFHW)

An EFHW is popular with portable operators because one antenna can cover multiple bands and often needs only one support point. When properly designed and installed, it can be quite efficient.

QUARTER-WAVE VERTICAL

A properly installed quarter-wave vertical with an effective radial or counterpoise system can be an excellent performer, especially on 20 and 40 meters. The return path is a major part of the antenna system.

More Efficient Antenna Options

FULL-SIZE VERTICAL

A full-size vertical can provide excellent efficiency while requiring relatively little horizontal space. With a good radial system and low-loss feed line, it can be a very effective portable or permanent antenna.

HEXBEAM

For operators who want directional gain, a Hexbeam can be an excellent option. A directional antenna can concentrate energy in a desired direction, increasing effective radiated power and improving received signal strength.

YAGI

A properly designed Yagi beam is one of the most effective choices when maximum performance is the goal. Its directional gain and front-to-back ratio allow an operator to concentrate available RF energy toward a specific area. The tradeoff is portability: a Yagi generally requires more equipment, support, and setup time.

Efficiency vs. Convenience

The *best* antenna isn't always the one with the highest theoretical efficiency. For a KPC portable operator, the best antenna may be the one that can be **deployed correctly, placed high enough, and used on the appropriate band.**

A well-installed dipole or EFHW can outperform a poorly installed 'better' antenna. Likewise, a small antenna that is convenient to deploy may produce more contacts simply because the operator can get it into the air quickly and effectively.

Helping Portable Operators in Less-Than-Ideal Conditions

Carry a Backup	A compact whip, short loaded vertical, or second wire cut for another band can keep an activation going when the primary antenna cannot be deployed.

Bring Radial / Counterpoise Wire	Pack enough wire for a practical return path. Keep radials and counterpoises as open as practical and away from people, metal structures, and other conductors.
Minimize Feed-Line Loss	Use the shortest practical feed line. Inspect coax, connectors, adapters, and strain relief before leaving home. Weatherproof exposed connections.
Match the Antenna to the Band	If the antenna must be installed lower than ideal, choose a band and mode that suit the resulting radiation pattern. One configuration will not perform equally well on every band.
Reduce Noise	Move away from power lines, electric fences, vehicles, solar equipment, and other noise sources when possible. Battery power can also reduce unwanted electrical noise.
Plan for Weather	Protect batteries and electronics from rain, condensation, dust, and direct sunlight. Cold weather can reduce battery capacity.
Test Before Deployment	Check the analyzer or SWR, tuner, feed line, output power, and the complete setup using the equipment you will actually carry.

The KPC Field Checklist

- [ ] Use an efficient antenna appropriate for the band.
- [ ] Get the antenna as high and clear as practical.
- [ ] Use an effective radial or counterpoise system.
- [ ] Minimize feed-line and connector losses.
- [ ] Carry a workable backup antenna configuration.
- [ ] Keep the antenna and radial system away from people and hazards.
- [ ] Choose the band and mode that fit current propagation.
- [ ] Listen for noise and openings before transmitting.
- [ ] Protect and test the complete portable station.

> **Solar minimum doesn't mean HF radio stops working.**
>
> **It means we have to become better operators. Optimize the station, adapt to the conditions, and make the contact.**

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73 — and see you on the air!